Complete Guide: Deploying a Flask Machine Learning API to Render

1. Project Structure

house_price_api/
■■■ train_model.py
■■■ app.py
■■■ house_model.pkl
■■■ requirements.txt
■■■ .gitignore

2. Train Model

Create train_model.py, train a Linear Regression model, and save it as house_model.pkl using joblib.

3. Flask API

Load the model using joblib.load('house_model.pkl'). Create '/' and '/predict' endpoints.

4. Install Packages

pip install flask pandas scikit-learn joblib gunicorn

5. Generate requirements.txt

pip freeze > requirements.txt

6. Local Testing

Run: python app.py and test http://127.0.0.1:5000

7. Git Commands

git init
git add .
git commit -m 'Initial Commit'
git branch -M main
git remote add origin https://github.com/<username>/house-price-api.git
git push -u origin main

8. Render Deployment

Create a Web Service, connect GitHub, use:
Build Command: pip install -r requirements.txt
Start Command: gunicorn app:app

9. Production Testing

Send POST requests with JSON {'size':1700} to your Render URL.

10. Common Error

Sending feature1/feature2 instead of size causes KeyError('size') and may lead to JSONDecodeError.

11. Update Deployment

```
git add .
git commit -m 'Updated API'
git push origin main
```